

EE 523: Digital VLSI Architecture Design

Course Code : EE 523

Course Name : Digital VLSI Architecture Design

L-T-P-C : 3-0-2-4.

Pre-requisite : IC161 - Applied Electronics, EE312P-Microelectronics Circuit Design Practicum, EE208P-Digital System-Design Practicum or Equivalent

Intended for :Final year BTech Electrical Engineering (EE), MS, M. Tech. & PhD.

Distribution : Elective for Final year B. Tech (EE), MS, M. Tech. & PhD

Approval: 14th Senate

Course Contents

- **Introduction:** Review of VLSI design flow, goals of VLSI design: optimization of speed, power dissipation, cost and reliability. [2 Lectures]
- **System Design Flow & Fixed-point Arithmetic:** Overview, system design flow, representation of numbers, floating point format, Qn.m format for fixed point arithmetic, floating-point to fixed-point conversion, block floating-point formats, forms of digital filters. [2 Lectures]
- **Algorithm to Architecture Transformation:** Architectural antipodes, transform approach to VLSI architectures, graph based formalism for describing processing algorithms, isomorphic architecture. [3 Lectures]
- **Equivalence Transforms for Combinational Computations:** Common assumptions, pipelining, replication, time sharing, associatively transform and other algebraic transforms. [5 Lectures]
- **Clocking of Synchronous Circuits:** Single-phase and two-phase clocking, wave pipelining, collective clock-buffer design, distributed clock-buffer trees, hybrid-clock distribution networks, impact of clock distribution delay on input/output (I/O) timing. [6 Lectures]
- **Asynchronous Data Processing Architectures:** Data consistency problem of vectored acquisition-plain bit parallel synchronization, unit distance coding, suppression of cross patterns, handshaking, partial handshaking, data consistency problem of scalar acquisition-synchronization at single place, synchronization at multiple places, synchronization from a slow clock, meta-stable synchronizer behavior. [5 Lectures]
- **Digital Signal Processing Using Array Architectures:** Systolic and wave-front arrays, mapping dependence and signal flow graphs to systolic and wave-front arrays, asynchronous communication protocols for wave-front arrays. [5 Lectures]
- **Architectural Synthesis and Optimization:** Circuit specifications for architectural synthesis, fundamental architectural synthesis problems, temporal domain-scheduling, spatial domain-binding, sequencing graphs, hierarchical models, synchronization problem, area and performance estimation, data path and control unit synthesis, constrained and unconstrained scheduling, scheduling of pipelined circuits. [8 Lectures]
- **CORDIC Based Architectures:** Introduction, CORDIC algorithm for hardware implementation, hardware mapping, time-shared architecture, C-slow time shared architecture, modified CORDIC algorithm,

recording of binary representation as 1, hardware optimization, optimal hardware design for CORDIC. [2 Lectures]

- **Digital Design of Communication Systems:** Top-level design options: bus-based design, point-to-point design, network-based design, hybrid connectivity, point-point KPN-based top-level design, KPN with shared bus and DMA controller, network-on-chip (NoC) top-level design, design of a router for NoC, run-time configuration, NoC for software defined radio. Typical digital communication system: source encoding, data compression, encryption, channel coding, framing, modulation, digital up-conversion and mixing, front end of the receiver. [4 Lectures]

Text Books:

1. Hubert Kaeslin, **Digital Integrated Circuit Design: From VLSI Architectures to CMOS Fabrication**, Cambridge University Press, 2009.
2. Giovanni De Micheli, **Synthesis and Optimization of Digital Circuits**, McGraw Hill, 2012.

References:

1. A.M. Fahim, **Clock generators for SoC processors: Circuits and Architectures**, Kluwer Academic Publishers, 2005.
2. Magdy A. Bayoumi, **VLSI Design Methodologies for Digital Signal Processing Architectures**, Springer, 2012.
3. Shoab Ahmed Khan, **Digital Design of Signal Processing Systems: A Practical Approach**, Wiley, 2011.
4. S.Y. Kung, **VLSI Array Processors**, Prentice Hall, 2012.